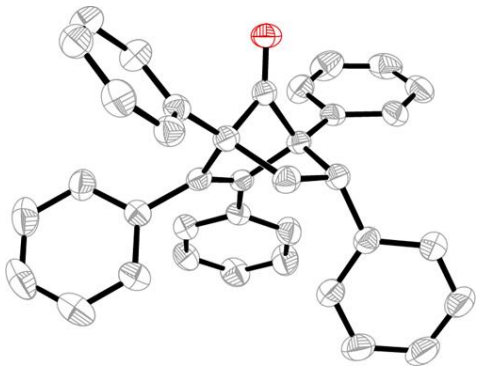
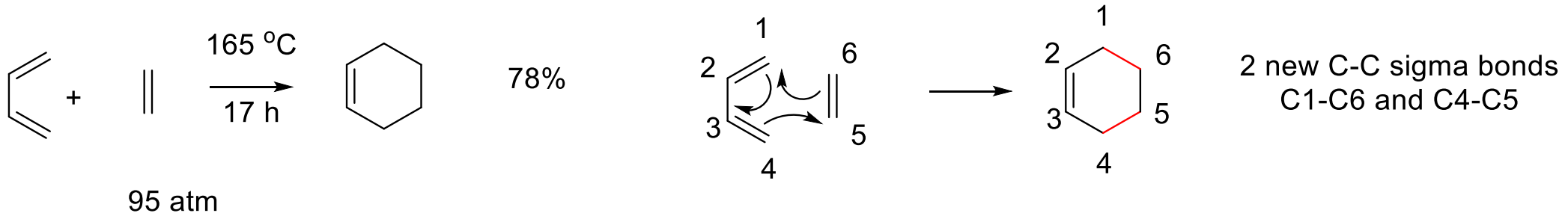
CM3191 Org L2: One-Pot Aldol Condensation and Diels–Alder Reaction



Dr. Hoang Truong Giang
chmhoan@nus.edu.sg

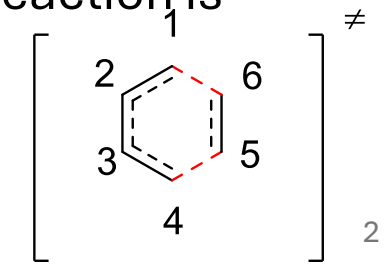
Diels-Alder reaction

- The Diels–Alder reaction is a reaction between a conjugated diene and an alkene to yield a cyclohexene ring. The simplest Diels–Alder reaction occurs between 1,3-butadiene and ethylene. The alkene partner is typically referred to as the dienophile.
- First reported in 1928; awarded the Nobel Prize in Chemistry in 1950.

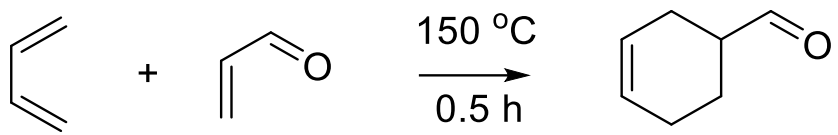


- Mechanism: one step, electrons move round a circle, no charge building up and no intermediates (classified as pericyclic reaction). Arrows can be drawn clockwise or anti-clockwise. All bond-forming and bond-breaking events occur simultaneously.

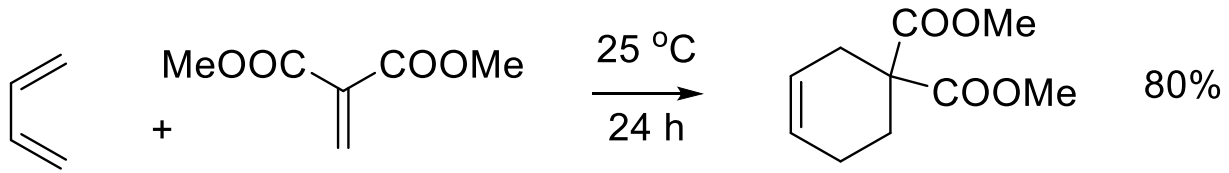
- In a DA reaction two C-C sigma bonds (C1-C6 and C4-C5) are formed,, at the expense of two C-C pi bonds. From energy point of view, Thermodynamically, the Diels–Alder reaction is favorable.

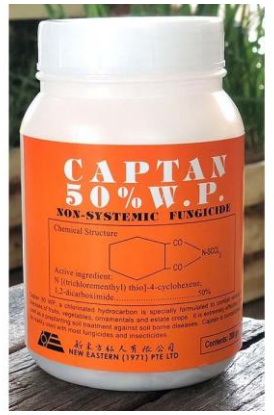


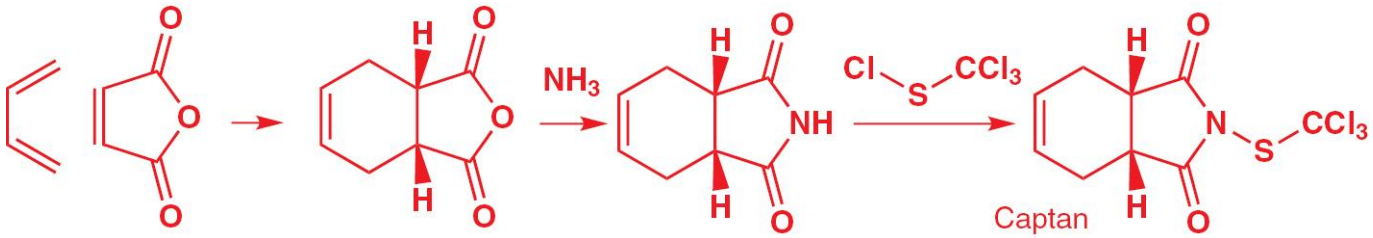
Examples of Diels-Alder reaction



The dienophile typically requires an electron-withdrawing group (EWG), or at least a phenyl (Ph) or chloro (Cl) substituent





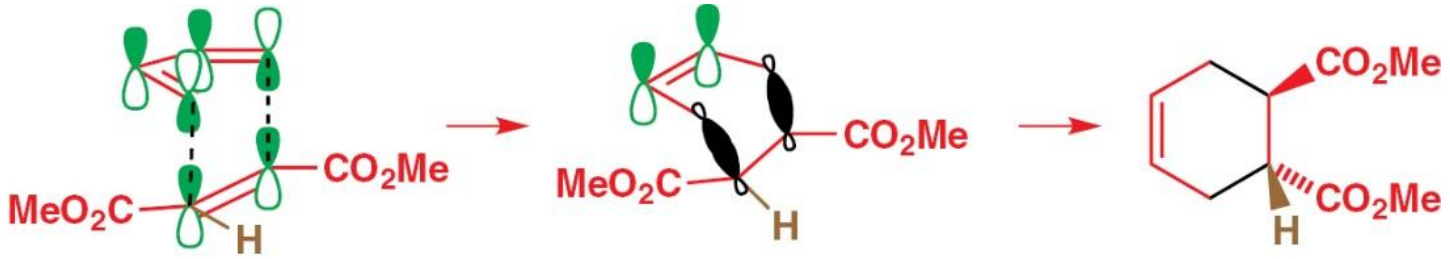


Captan is a non-systemic phthalimide fungicide used extensively in agriculture to control a broad range of fungal diseases, such as scab and fruit rot, on various fruits, vegetables, and ornamental crops.

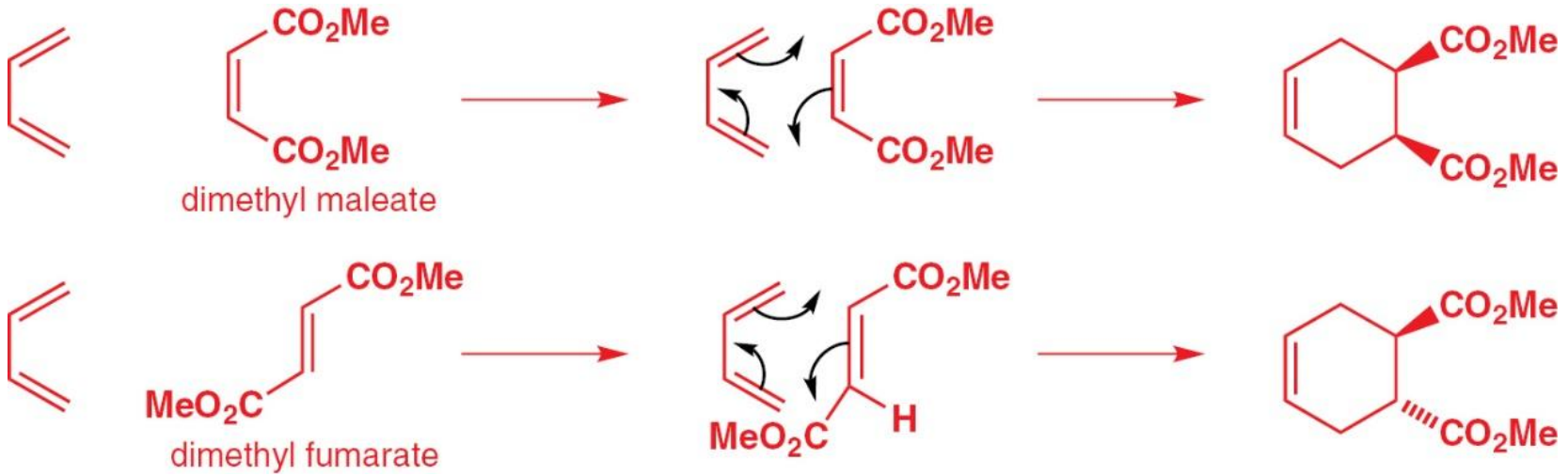
Photo from: https://shopee.sg/Fungicide-Non-Systemic-Captan-50-WP-Plant-Indoor-Outdoor-Garden-i.331190305.9535406571

Stereoselectivity: dienophile

- Because the Diels–Alder mechanism is concerted without bond rotation, the reaction is stereospecific

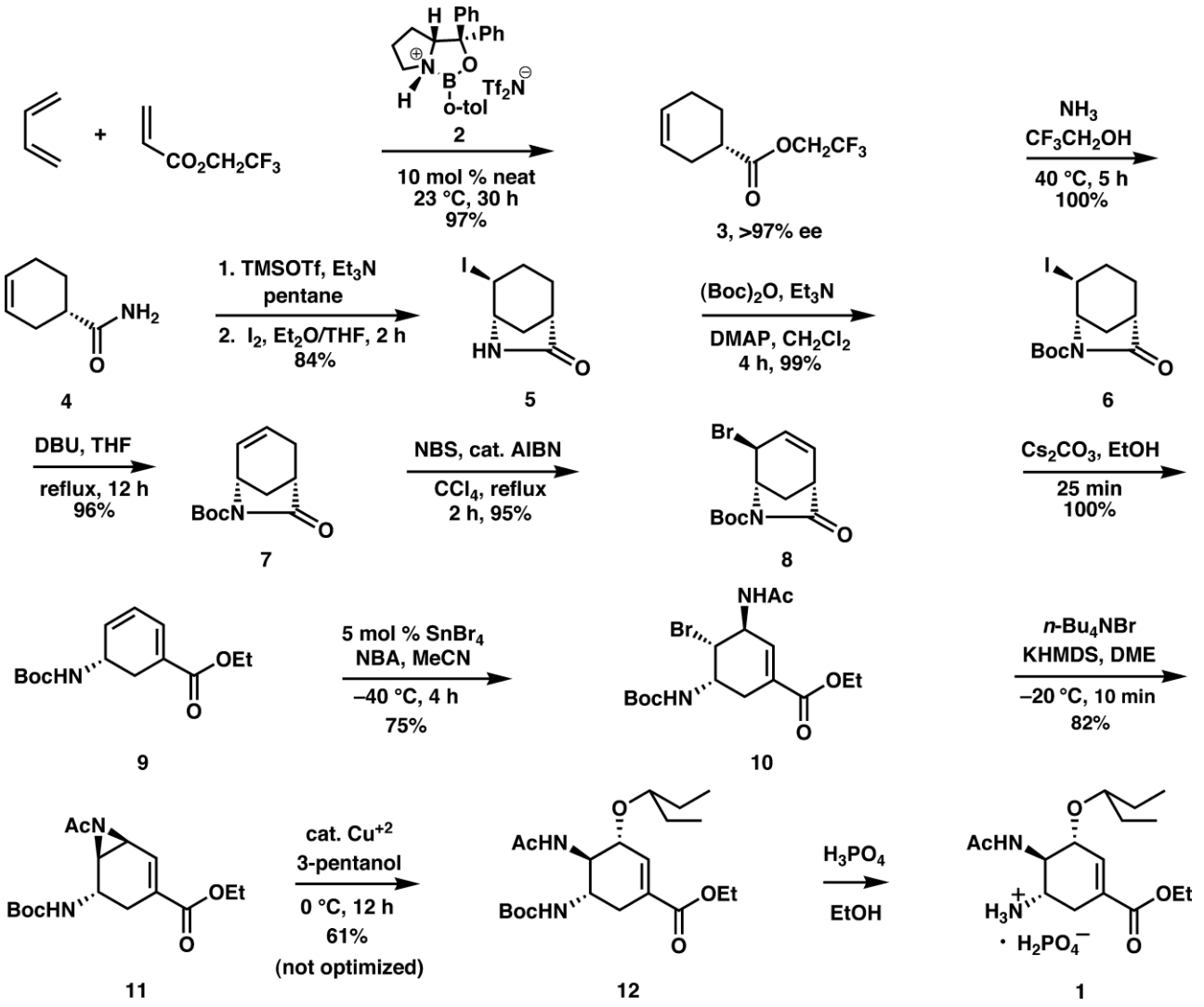


- A *cis*-dienophile yields a *cis*-product; a *trans*-dienophile yields a *trans*-product.



Diels – Alder in pharmaceutical synthesis

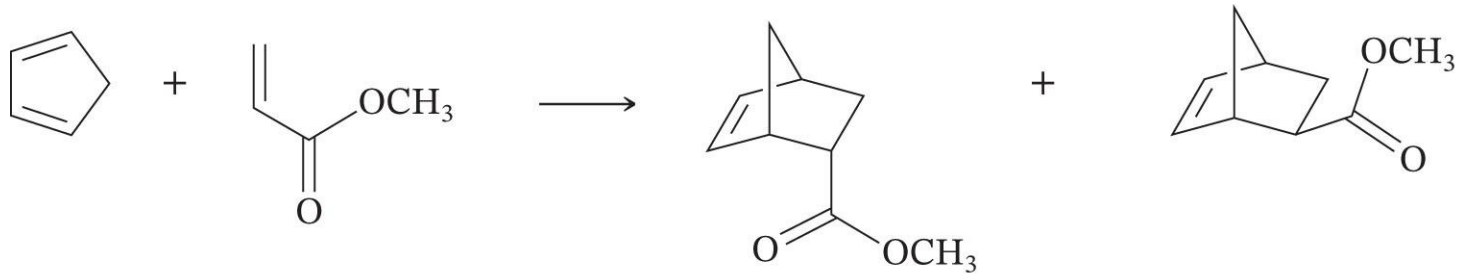
Tamiflu: Antiviral medication used for the prevention and treatment of influenza A and B.



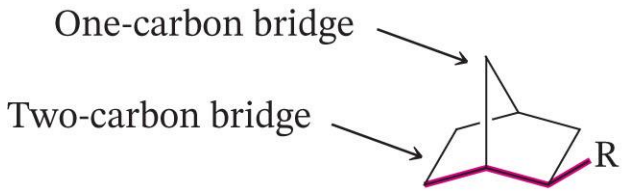
Corey et al, A Short Enantioselective Pathway for the Synthesis of the Anti-Influenza Neuramidase Inhibitor Oseltamivir from 1,3-Butadiene and Acrylic Acid Journal American Chemical Society. 2006;128(19):6310-6311. doi:10.1021/ja0616433

Stereochemistry: endo vs exo

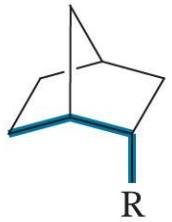
- When a cyclic diene reacts with an unsymmetrical dienophile, two products are possible



- Endo and exo



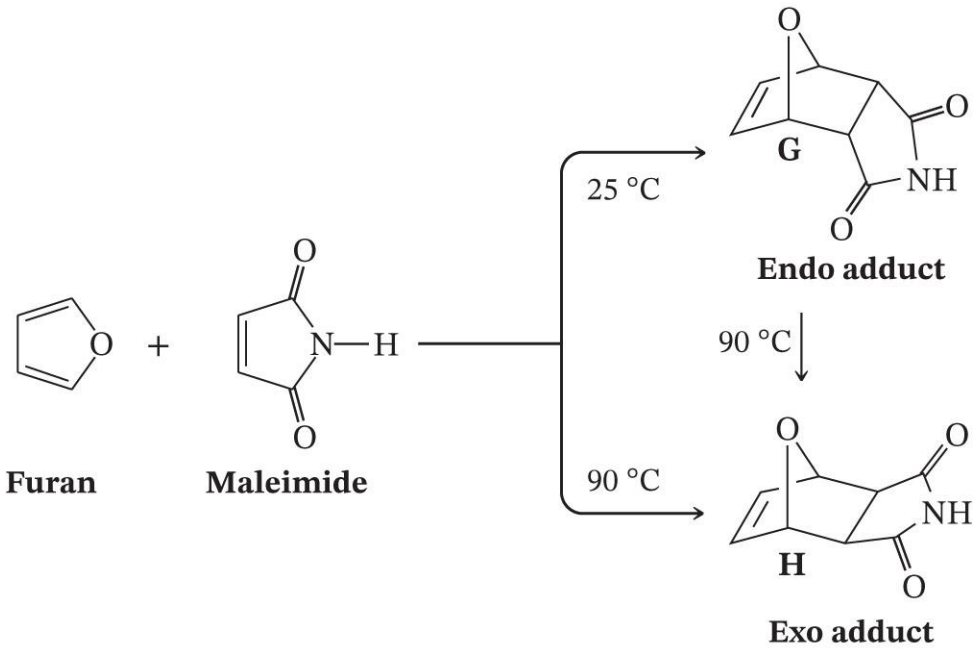
R is exo
(oriented away from the longest bridge)



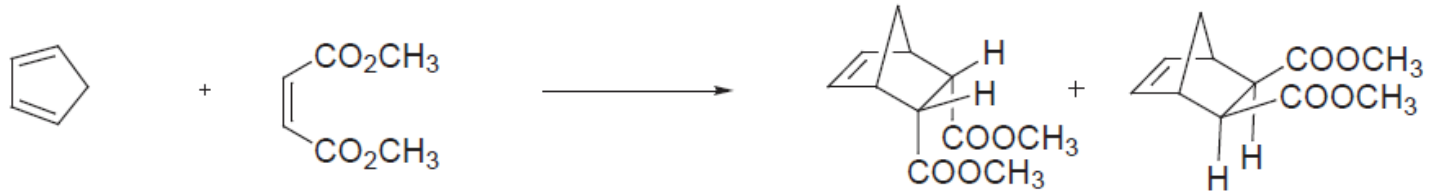
R is endo
(oriented closer to the longest bridge)

Endo vs exo

- The Diels-Alder reaction occurs primarily in an endo rather than exo fashion when the reaction is kinetically controlled (lower temperature).
-
- When the Diels-Alder reaction is carried out at a higher temperature, the major product is the exo-isomer.



Endo and exo products

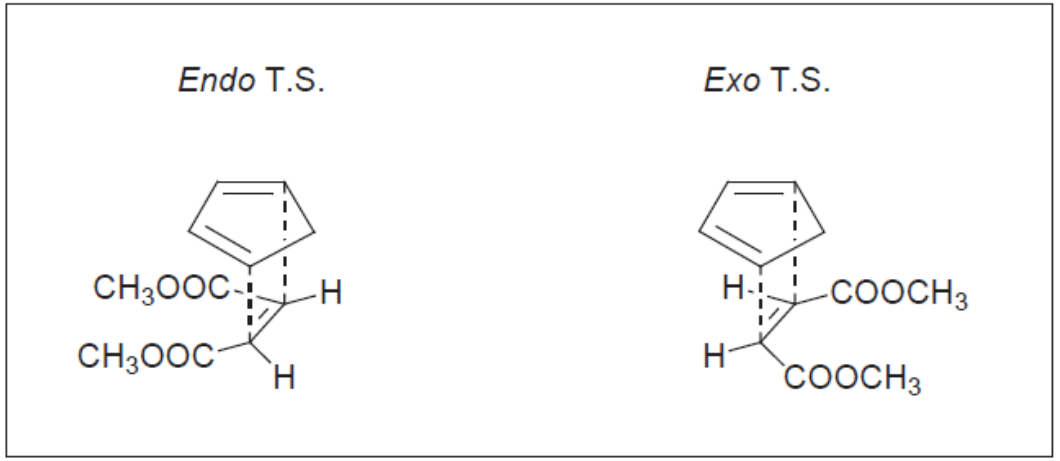


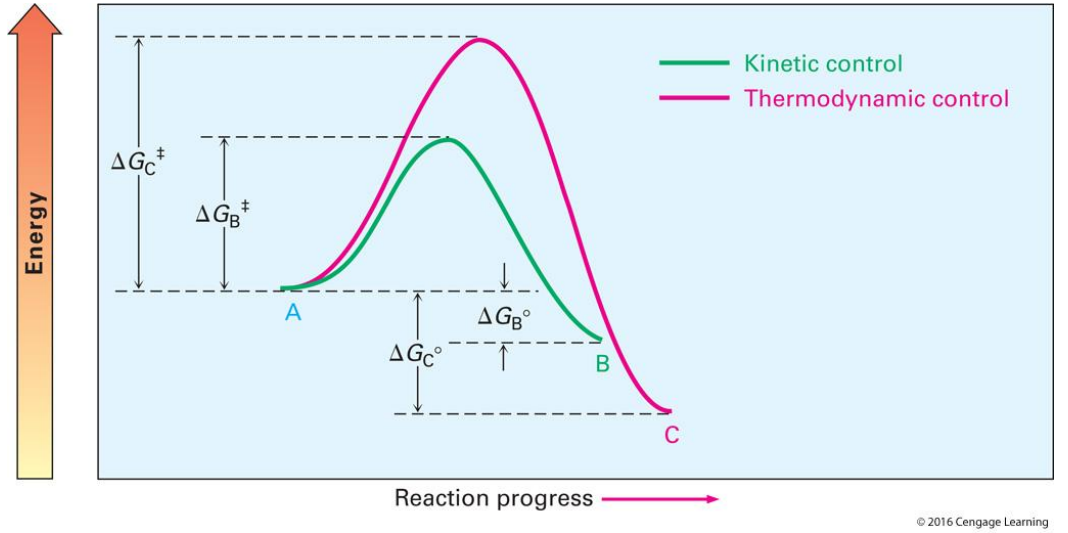
endo

Major

exo

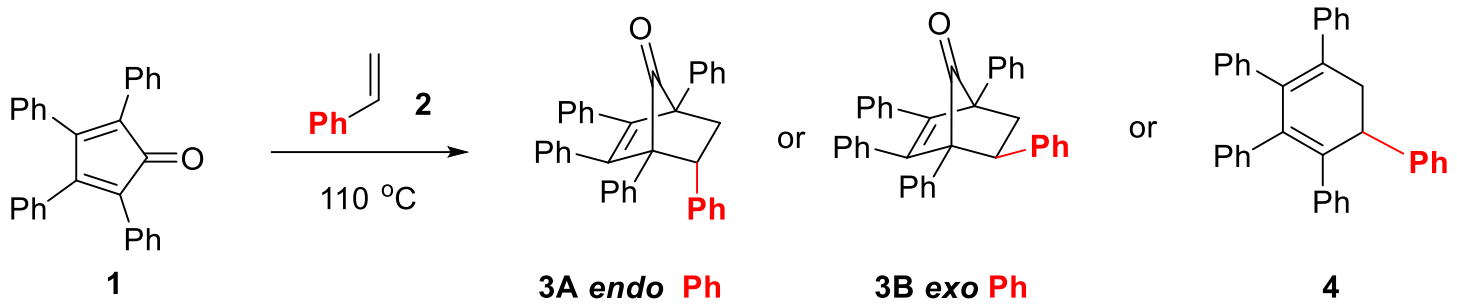
Minor





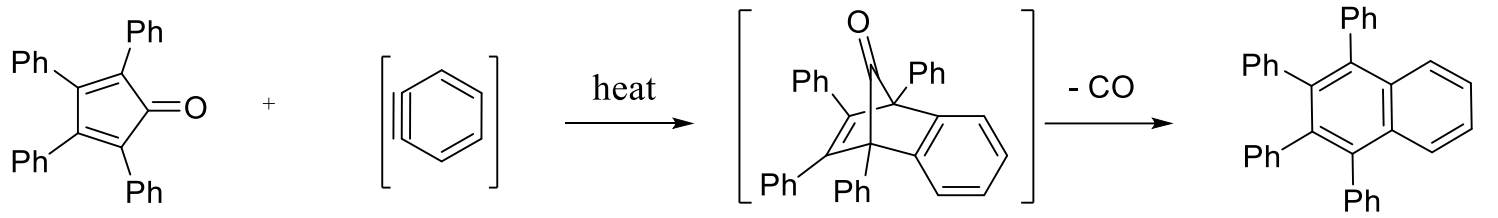
The *endo* adduct forms faster but is thermodynamically less stable (kinetic product).

Experiment 2 Diels-Alder reaction: endo vs exo



Literature for decarbonylation reaction

Reaction with benzyne



only product isolated

Reaction with 1,2-diphenylacetylene

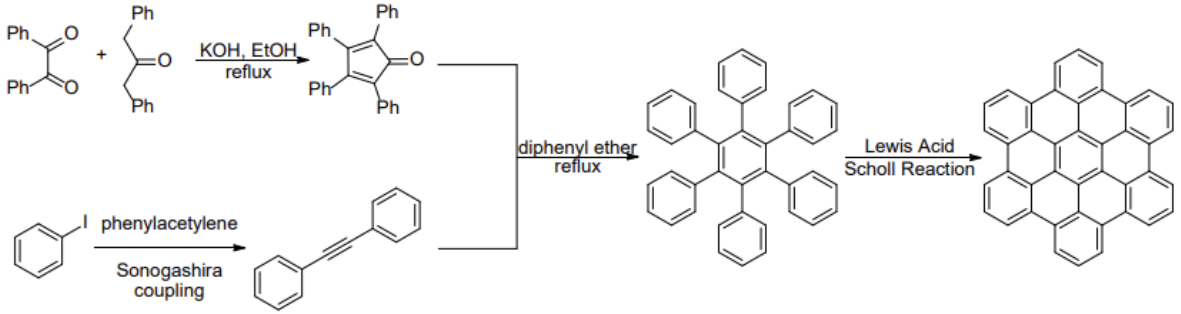
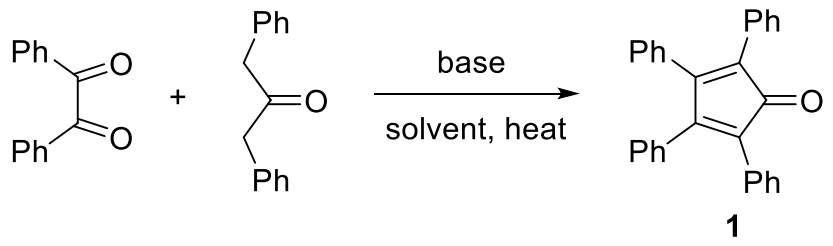
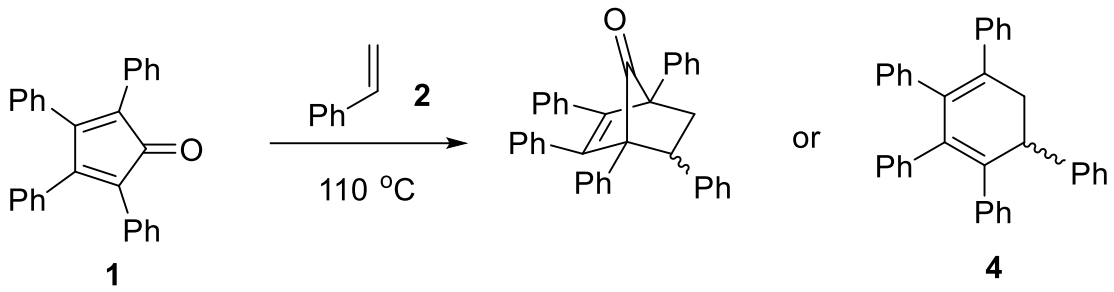
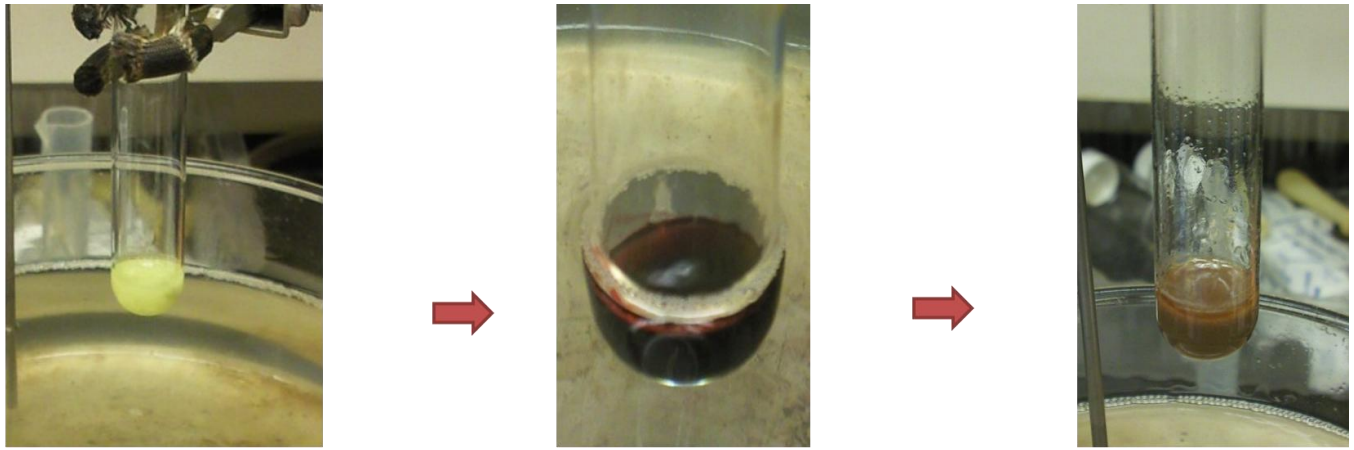


Figure 1 - Synthetic route to hexa-peri-hexabenzocoronene

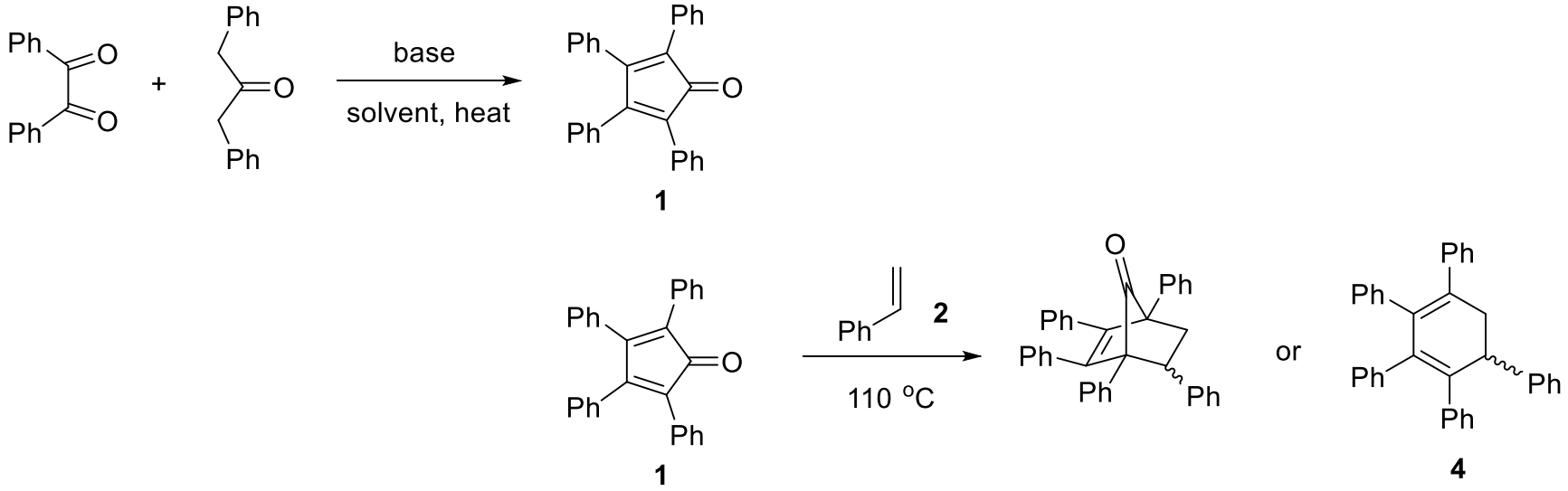
One-pot aldol condensation/Diels-Alder reaction







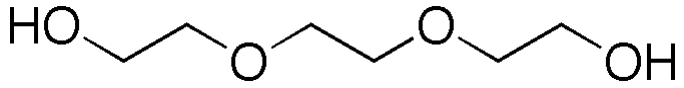
One-pot Aldol condensation/Diels-Alder reaction



Advantage of one-pot reaction

- Reduces solvent consumption and materials required to isolate intermediates
- Reduces the number of reaction vessels required in industrial processing.
- Traps reactive or unstable intermediates *in situ*.

Triethylene glycol



Negligible Volatility

With an extremely low vapor pressure (<0.01 mmHg at 20°C) and high boiling point (287°C), TEG eliminates toxic inhalation exposure and solvent evaporation during heating.

High Thermal Stability

TEG withstands sustained temperatures exceeding 110°C without thermal degradation, permitting rapid condensation and concerted cycloaddition in a single heated vial.

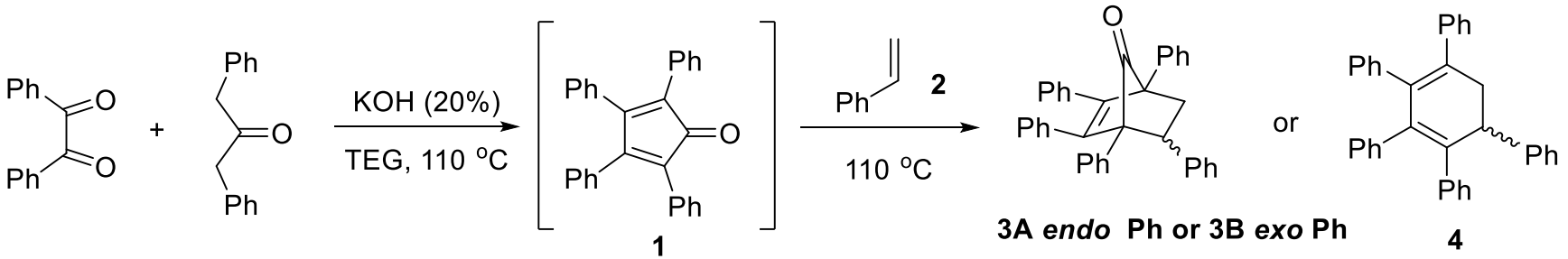
Favorable Toxicity Profile

TEG exhibits low toxicity and readily undergoes aerobic biodegradation into harmless CO_2 and H_2O in wastewater systems.

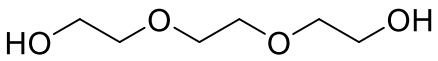
https://en.wikipedia.org/wiki/Triethylene_glycol

Learning objectives

- Synthesize: Execute a one-pot Aldol condensation and Diels–Alder reaction.
- Determine the major regio- and stereochemical outcome of the Diels–Alder reaction using spectroscopic methods.

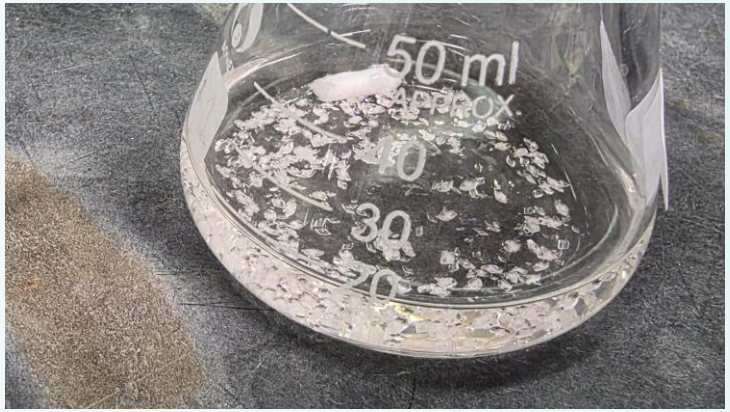


TEG = triethylene glycol



3A, **3B**, and **4** would each be enantiomeric mixtures

Laboratory Operational Workflow



Step 1: Dissolution

Combine benzil (0.21 g), 1,3-diphenylacetone (0.21 g), and 0.5 mL TEG. Warm to 110°C until a homogeneous yellow melt forms.

Step 2: Aldol condensation

Introduce ethanolic KOH. Observe immediate shift to dark purple tetracyclone. Heat 3–5 min and confirm completion via R1 TLC.

Step 3: [4+2] Reaction

Charge 0.40 mL styrene directly. Heat at 110°C for 30 min until dark purple fades to tan. Verify disappearance of diene via R2 TLC.

Step 4: Isolation

Cool to RT, quench with ethanol, and chill on ice to precipitate adduct. Collect via Hirsch funnel vacuum filtration and recrystallize.